SOLUTIONS: DISCRETE EVENT SYSTEMS
MASTER IN CONTROL

1. Exercise

a) We flag initially (with a plus) all pair of states that are terminal and non-terminal. This is done according to the following table:

1	+			
2		+		
3	+		+	
4	+		+	
	0	1	2	3

Then, all state pairs that have different sets of enabled events are flagged with an x. This is done according to the following table:

1	+			
2		+		
3	+		+	
4	+	x	+	x
	0	1	2	3

Finally, we need to flag with an o those pair of states such that an event enabled in both states leads to a pair of states which are already flagged. As there are no such pairs of states, unlabeled pairs of states are equivalent to each other. In particular, the equivalence classes are: $\{0,2\},\{1,3\},\{4\}$.

Some students did not list $\{4\}$ as an equivalence class. The fact that it contains a single element does not mean that it is not an equivalence class.

b) To find $G_{\min}$ it is enough to identify in G those states that are equivalent. In other words, the states of the minimal automaton are the equivalence classes of the states of the original automaton G, as shown in Fig. 1.1.

c) The automaton implementing the parallel composition $G_1||G_2$ is shown in Fig.

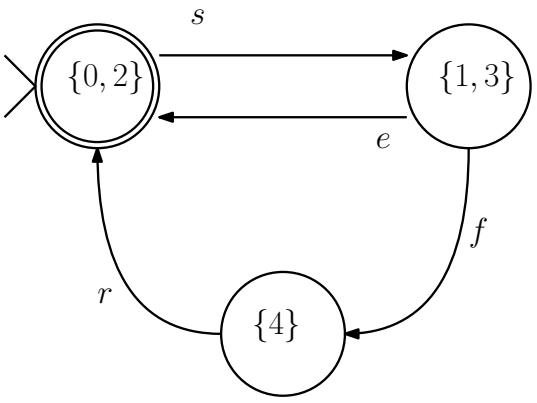


Figure 1.1 Transition diagram of minimal automaton $G_{\min}$

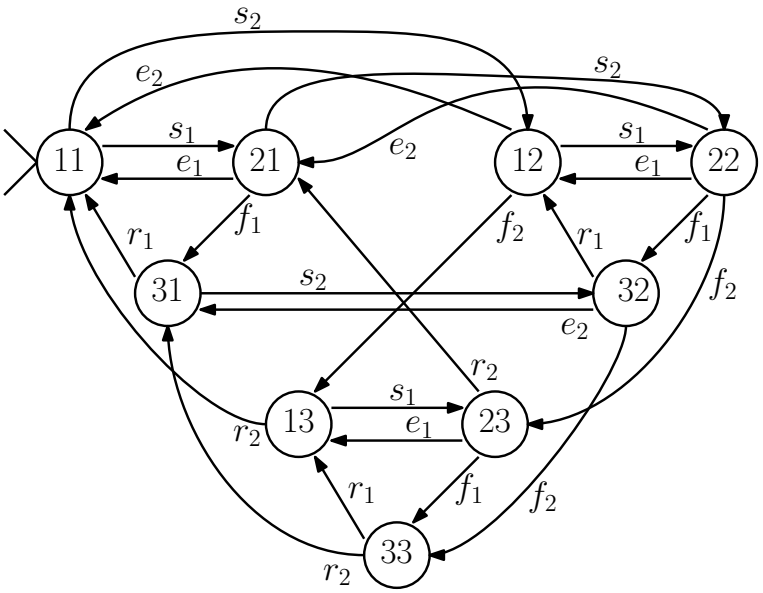


Figure 1.2 Transition diagram of $G_1||G_2$

1.2.

d) Projection over the subalphabet $F = \{f_1, f_2, r_1, r_2\}$ can be achieved in a non-deterministic automaton by replacing all events in $E \backslash F$ with ε event. The corresponding automaton G_N is shown in Fig. 1.3.
Some students erroneously removed transitions including events not contained in F from the state transition diagram. This is, of course, very different from replacing such events with the ε event and completely alters the end result.

e) Finally, the observer automaton $Obs(G_N)$ is sketched in Fig. 1.4

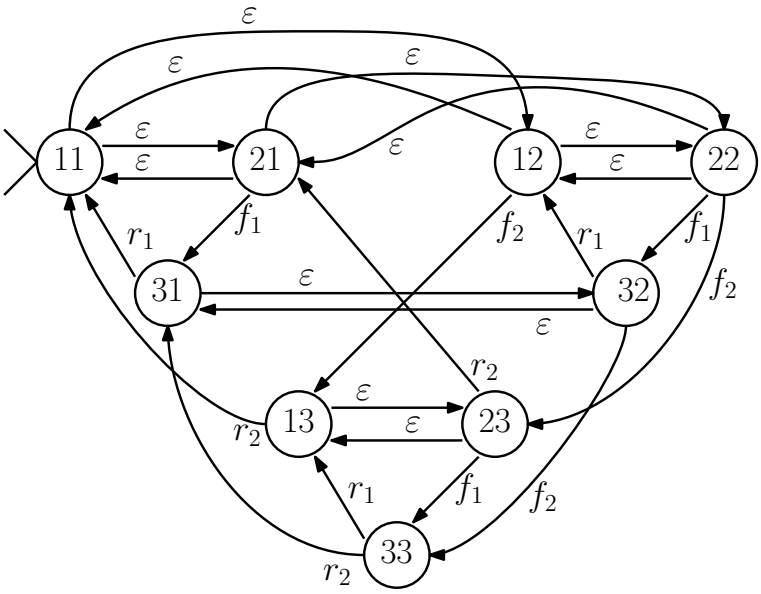


Figure 1.3 Transition diagram of G_N

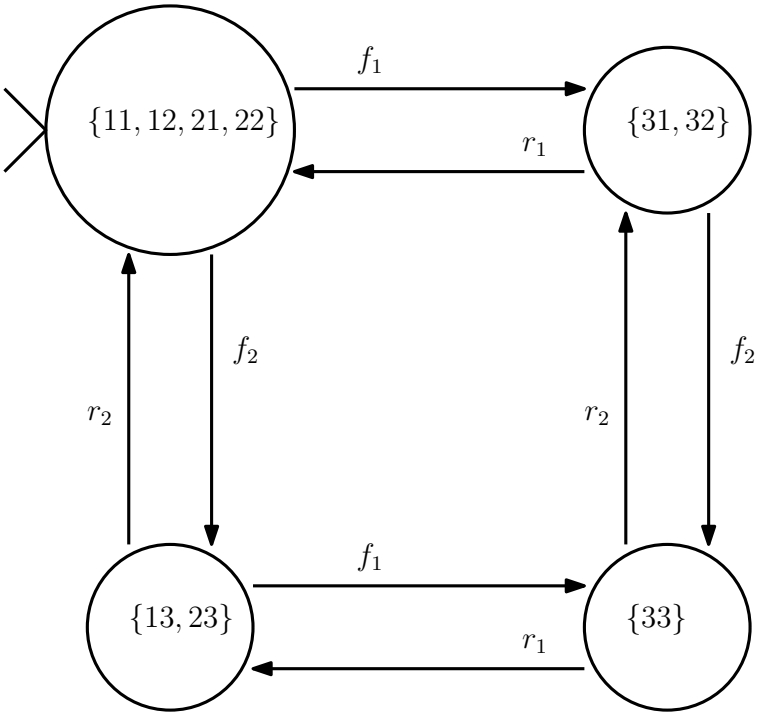


Figure 1.4 Observer automaton

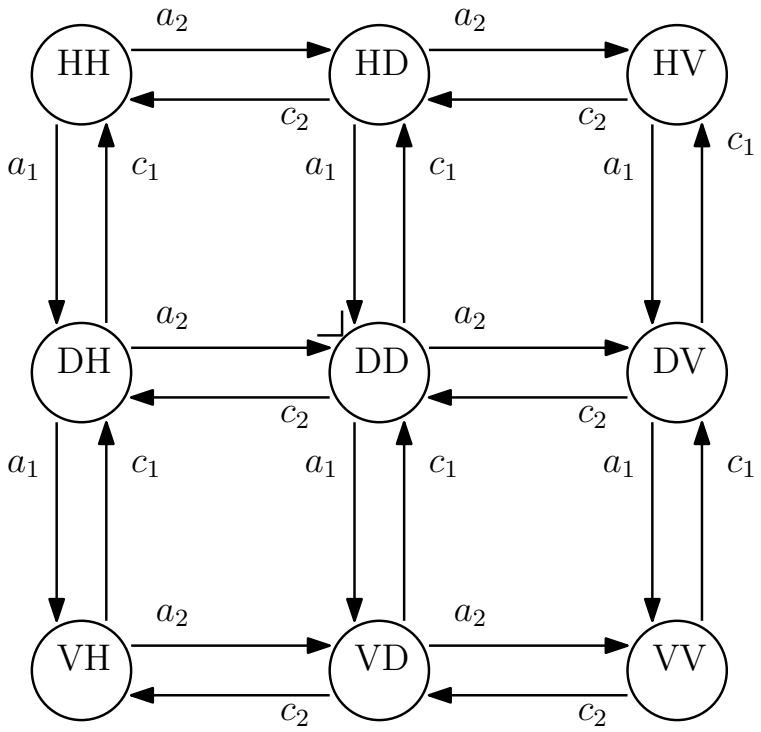


Figure 2.1 Automaton G, modeling the 2 deflecting mirrors

2. Exercise

a) We build an automaton G with event set $E = \{c_1, a_1, c_2, a_2\}$ and state-space $X = \{HH, HD, HV, DH, DD, DV, VH, VD, VV\}$ as shown in Fig. 2.1.
Several students missed a few states in X. This happens, normally, when the states are not arranged in a clear pattern as in Fig. 2.1. Keeping your diagrams simple will make it much easier to also make sure they are correct.

b) There are several mirrors configurations in which light is bounced back towards the Light Source. In particular, if mirror 1 is horizontal, this directly bounces back the light towards the light source. Also if mirror 1 is diagonal and mirror 2 is vertical light bounces back towards the light source. As a consequence the specification can be modeled as a forbidden states specification, only allowing states $\{DD, DH, VH, VD, VV\}$. The corresponding automaton is shown in Fig. 2.2.

c) Concurrent composition $G_K||G$ yields an automaton which is isomorphic to G_K, and where state labels are each time repeated twice. See Fig. 2.3 where dashed arrows are used to emphasize disabled events. It can be seen that only c_1 and a_2 events are disabled. Hence, when $E_{uc} = \{c_2\}$ the specification $\mathscr{L}(G_K)$ is controllable.

d) The same analysis shows that in order to maximize the set of uncontrollable events that still keep the specification controllable one can take $E_{uc} = \{c_2, a_1\} = E \backslash \{c_1, a_2\}$.

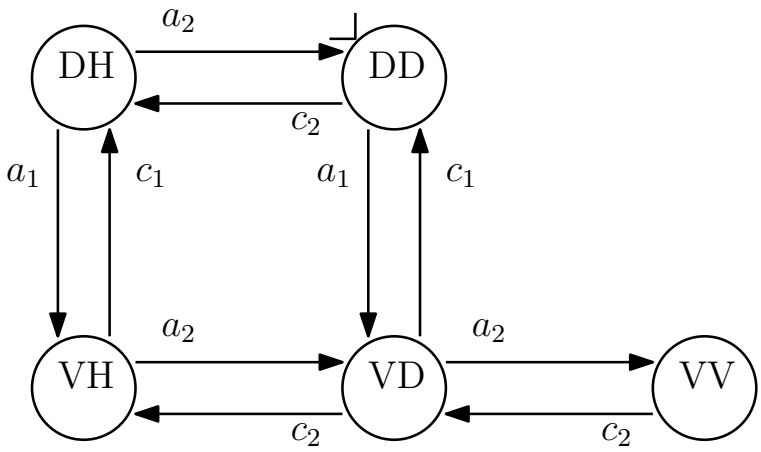


Figure 2.2 Automaton G_K modeling the specification

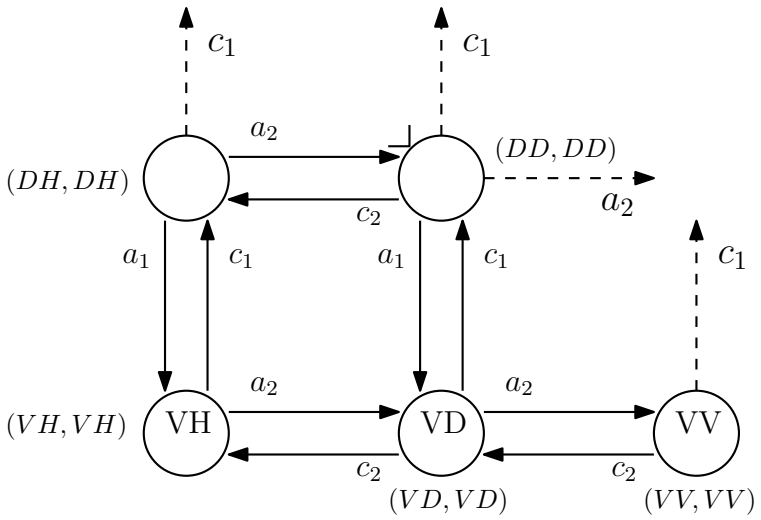


Figure 2.3 Composition $G||G_K$ and disabled events

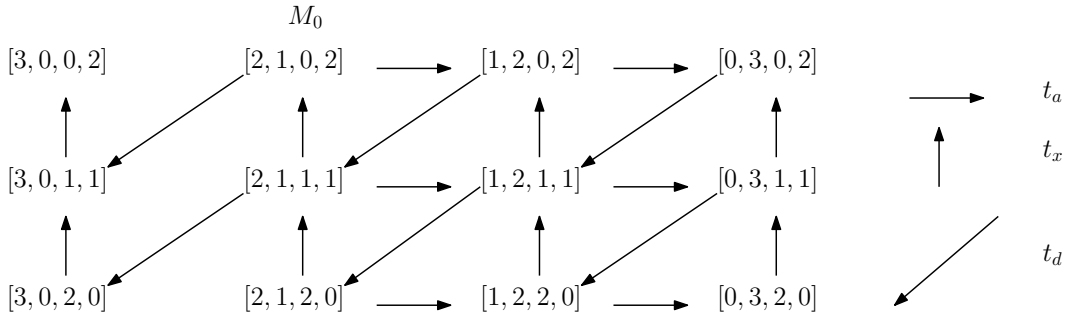


Figure 3.1 Transition diagram and reachable set $\mathscr{R}(M_0)$

3. Exercise

a) The matrices *Pre* and *Post*, ordering places as p_1, p_2, p_3, p_4 and transitions as t_a, t_d, t_x are given below:

$$Pre = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \qquad Post = \begin{bmatrix} 0 & 1 & 0 \\ 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The incidence matrix C is computed as follows:

$$C = Post - Pre = \begin{bmatrix} -1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}.$$

b) The tokens in p_2 model the total number of people in Room 1, including the guardian. Notice that at least 1 person (the guardian) must be in place p_2 for a new arrival to occur. On the other hand, when t_a occurs, p_1 decreases by 1. This means that p_1 is modeling the residual capacity of Room 1. When t_d occurs the number of tokens increases by 1, while it decreases when t_x occurs. p_3 is therefore modeling the number of people in Room 2. Accordingly p_4 is measuring the residual capacity of Room 2.

c) There are two minimal P-semiflows, the vectors: $[1,1,0,0]$ and $[0,0,1,1]$. These correspond to conservation of Room capacity + Total number of people (possibly including the guardian) for the two rooms.

d) There is one minimal T-semiflow for the network, the vector $[1,1,1]'$. This corresponds to the fact that any person arriving, transiting to the second room and then exiting the exhibition leaves the current state of the Net unchanged.

e) The reachable set from marking M_0 is sketched in Fig. 3.1.
Several students missed a few states from the reachable set. This happens, normally, when the states are not arranged in a clear pattern as in Fig. 3.1. Keeping your diagrams simple will make it much easier to also make sure they are correct.

f) The Petri Net is structurally bounded as $[1,1,1,1]$ is a P-semiflow and its support is the whole set of places P. For the same reason the network is structurally conservative.

g) The Petri Net (N, M_0) is not reversible as there exist markings in $\mathscr{R}(M_0)$ from where it is no longer possible to reach the initial states. For instance, the marking $M = [3,0,0,2]$.

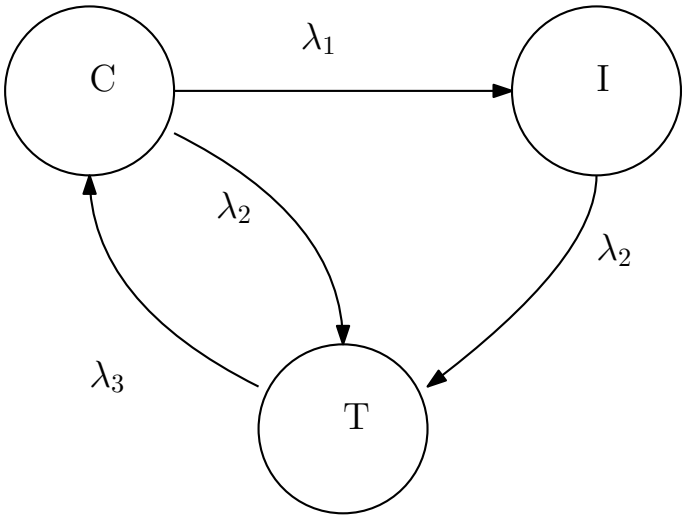


Figure 4.1 Transition diagram of the Markov Chain

4. Exercise

a) The transition diagram of the Markov Chain describing the fleet of EVs is shown in Fig. 4.1.

b) Yes, the Markov Chain is ergodic as its transition diagram is strongly connected (as a graph).
Several students misunderstood the notion of absorbing strongly connected component. When there is a unique strongly connected component this is also trivially absorbing.

c) Denoting π_1, π_2 and π_3 the probabilities of being in states C, I and T respectively, and by π the row vector $[\pi_1, \pi_2 \pi_3]$ yields the following system of equations:

$$\dot{\pi} = \pi Q \qquad Q = \begin{bmatrix} -(\lambda_1+\lambda_2) & \lambda_1 & \lambda_2 \\ 0 & -\lambda_2 & \lambda_2 \\ \lambda_3 & 0 & -\lambda_3 \end{bmatrix}.$$

d) To compute the steady-state probability distribution we must solve the following system of linear equations:

$$\begin{cases} \pi_s Q = & 0 \\ \pi_{1s} + \pi_{2s} + \pi_{3s} & = 1 \end{cases}$$

In particular then:

$$\lambda_3 \pi_{3s} = \lambda_2(\pi_{1s} + \pi_{2s}) = \lambda_2(1 - \pi_{3s}).$$

This yields: $\pi_{3s} = \frac{\lambda_2}{\lambda_2+\lambda_3}$. Substituting in the other equations yields, after straightforward calculations:

$$\pi_{1s} = \frac{\lambda_2 \lambda_3}{(\lambda_2+\lambda_3)(\lambda_1+\lambda_2)},$$

$$\pi_{2s} = \frac{\lambda_1 \lambda_3}{(\lambda_2+\lambda_3)(\lambda_1+\lambda_2)}.$$

e) The coefficient λ_1 is proportional to charging speed. Notice that $\frac{1}{\lambda_1}$ is the average time it would take to charge the battery if no early departures would be allowed. Denote by $\chi_3(x)$ the characteristic function of state 3, namely $\chi_3(x) = 1$

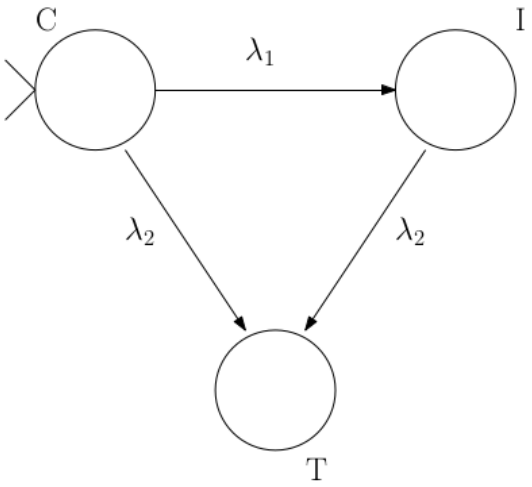


Figure 4.2 Time between journeys: associated Markov chain

when $x = 3$ and 0 otherwise. Computing the average time spent travelling we have, by virtue of the ergodic theorem:

$$\lim_{T\to+\infty} \frac{\int_0^T \chi_3(x(t))\,dt}{T} = \sum_{i=1}^{3} \pi_{is}\chi_3(x_i) = \pi_{3s},$$

with probability 1. Notice that $\pi_{3s} = \frac{\lambda_2}{\lambda_2+\lambda_3}$, and indeed $\partial\pi_{3s}/\partial\lambda_1 = 0$.
Some students missed the simplification of the steady-state probability expression. This prevented them from finding a correct justification of the claim.

f) After a journey the electric vehicles enters the state C. The Markov chain describing the states visited between consecutive journeys is therefore as sketched in Fig. 4.2. In order to answer the question we need to compute the average absorption time of state T from state C. Since transition to I does not affect the transition rate towards T, the average absorption time is $1/\lambda_2$.